UK Air Quality 2025: SQL Function Analysis

Nov 4, 2026 · @Mustafa Malik

1. Perspective

Project type: Type 3. There are 15 questions, and each one is built around a different family of SQL functions.

Perspective: Air pollution exposure, measured against UK air quality objectives and the WHO 2021 health guidelines.

End consumer: A local authority air quality or public health officer. They need to know where concentrations are highest, which site types and regions are most affected, when pollution peaks, and how far each area is from the targets it is working towards.

Benchmarks used:

Pollutant	UK annual objective	England target	WHO 2021 annual guideline
NO2	40 µg/m³	–	10 µg/m³
PM10	40 µg/m³	–	15 µg/m³
PM2.5	20 µg/m³	10 µg/m³ by 2040	5 µg/m³
O3	Daily max 8-hour mean above 100 µg/m³ on no more than 10 days a year	–	–

Daily PM2.5 is also banded with the UK Daily Air Quality Index (DAQI): Low 0–35, Moderate 36–53, High 54–70, Very High 71+ µg/m³.

2. Dataset

Source: Defra UK-AIR, Automatic Urban and Rural Network (AURN), the UK's national network of automatic air quality monitors. One pre-formatted hourly file was downloaded per site for 2025. 201 of the network's 215 sites had a 2025 file.

Preparation: The per-site files are wide. Each site has its own set of pollutant columns, and Marylebone Road alone has about 50. A PowerShell script reshaped them into one long file with a row per site, hour and pollutant, keeping NO2, O3, PM10 and PM2.5. A second script read each site's Defra page to get its environment type, region and coordinates.

Database: `air_quality` (MySQL 8)

Table	Rows	Contents
measurements	4,516,584	site_code, obs_date, obs_hour (1–24, hour ending), measured_at (start of the hour), pollutant, concentration (µg/m³), status
sites	201	site_code, site_name, environment_type, region, nation, latitude, longitude
v_site_annual (view)	–	annual mean and data capture per site and pollutant

Coverage: 1 January to 31 December 2025, GMT, all readings ratified.

- NO2: 155 sites
- O3: 97 sites
- PM10: 146 sites
- PM2.5: 158 sites

The sites are 88 urban background, 78 urban traffic, 24 rural background and 11 industrial or suburban. By nation, 159 are in England, 23 in Scotland, 11 in Wales and 8 in Northern Ireland.

Conventions:

- Annual means use sites with at least 75% data capture (6,570 of 8,760 hours), the usual Defra threshold.
- Daily means need at least 18 hours of data.
- Small negative readings are instrument noise around zero.

3. Core Functions (Q1–Q8)

Aggregate, grouping, string, pattern-matching, numeric, conditional and date functions.

Q1: Aggregate functions

Question: For each pollutant, how many hourly readings and sites are there, what are the mean, lowest and highest readings, and what is the average data capture?

Concept: COUNT, COUNT(DISTINCT), AVG, MIN, MAX, SUM

pollutant	readings	sites	mean_ugm3	min_ugm3	max_ugm3	negative_readings	avg_capture_pct
NO2	1264600	155	15.57	-0.3	172.2	7	93.1
O3	792219	97	54.36	-0.3	199.4	20	93.2
PM10	1190148	146	13.65	-3.9	4964.7	578	93.1
PM25	1269617	158	7.99	-5.0	670.5	1999	91.7

4 rows returned (shown in full).

Interpretation: The network recorded 4.52 million hourly readings in 2025, with average data capture of 91.7–93.2% for every pollutant. Mean concentrations were 15.6 µg/m³ for NO2, 54.4 for ozone, 13.7 for PM10 and 8.0 for PM2.5. NO2, PM10 and PM2.5 are all well below their UK annual objectives, but NO2 and PM2.5 are above the WHO guidelines (10 and 5). The PM10 maximum of 4,964.7 is a single extreme spike, most likely an instrument or local event, so later questions rely on means rather than maxima. Negative readings are rare (2,604 in total, mostly PM2.5) and are noise around zero.

Q2: Grouping functions

Question: What is the average annual NO2 concentration by nation and site type, with a subtotal for each nation and a UK total?

Concept: GROUP BY … WITH ROLLUP + GROUPING() to label the subtotal and total rows

nation	site_type	sites	mean_no2_ugm3	highest_site_ugm3	sites_above_uk_40	sites_above_who_10
England	Rural Background	8	6.1	10.7	0	1
England	Suburban Background	2	11.4	16.4	0	1
England	Suburban Industrial	2	14.1	14.4	0	2
England	Urban Background	50	14.1	24.8	0	43
England	Urban Industrial	3	15.6	21.7	0	3
England	Urban Traffic	46	20.5	30.6	0	44
England	(nation total)	111	16.2	30.6	0	94
Northern Ireland	Urban Background	1	8.1	8.1	0	0
Northern Ireland	Urban Traffic	2	27.4	33.5	0	2
Northern Ireland	(nation total)	3	21.0	33.5	0	2
Scotland	Rural Background	2	2.5	3.4	0	0

nation	site_type	sites	mean_no2_ugm3	highest_site_ugm3	sites_above_uk_40	sites_above_who_10
Scotland	Suburban Background	1	6.9	6.9	0	0
Scotland	Urban Background	5	10.1	14.6	0	3
Scotland	Urban Industrial	2	11.6	12.2	0	2
Scotland	Urban Traffic	8	19.0	34.8	0	8
Scotland	(nation total)	18	13.2	34.8	0	13
Wales	Rural Background	2	3.0	3.0	0	0
Wales	Urban Background	3	11.7	14.0	0	2
Wales	Urban Industrial	1	9.3	9.3	0	0
Wales	Urban Traffic	4	18.8	22.3	0	4
Wales	(nation total)	10	12.5	22.3	0	6
ALL NATIONS	(UK total)	142	15.6	34.8	0	115

22 rows returned (shown in full).

Interpretation: No site in any nation exceeded the UK annual NO2 objective of 40 µg/m³, but 115 of the 142 sites with enough data (81%) were above the WHO guideline of 10. Roadside (Urban Traffic) sites were highest everywhere: 20.5 in England, 19.0 in Scotland, 18.8 in Wales and 27.4 at Northern Ireland's two roadside sites. Rural background sites averaged only 2.5–6.1. 58 of the 60 roadside sites exceeded the WHO guideline, so the gap to health-based levels is mainly a traffic problem.

Q3: String functions: extraction

Question: Site names start with the town or city and end with a descriptor ("London Marylebone Road", "Belfast Centre"). Which towns have three or more sites, what descriptors do they use, and what is their average annual NO2?

Concept: CTE + SUBSTRING_INDEX (first and last word) + GROUP_CONCAT(DISTINCT ...)

town	sites	descriptors	sites_with_no2	mean_no2_ugm3
London	12	Bexley, Bloomsbury, Harlington, Hillingdon, Kensington, Park, Road, School, South, Street, Westminster	8	20.9

town	sites	descriptors	sites_with_no2	mean_no2_ugm3
Birmingham	4	Green, Ladywood, Roadside, University	3	18.5
Glasgow	4	Kerbside, Road, Street, Townhead	4	21.0
Bristol	3	Patchway, Paul's, Way	2	20.7
Nottingham	3	Bilborough, Boulevard, Gardens	2	17.5
Reading	3	Lane, Road, Town	2	14.8
Sheffield	3	Green, Road, Tinsley	3	21.1

7 rows returned (shown in full).

Interpretation: London has the most monitoring sites (12, of which 8 have a full year of NO2), averaging 20.9 µg/m³. Among towns with three or more sites, Sheffield (21.1) and Glasgow (21.0) have the highest averages and Reading the lowest (14.8). The descriptor column shows the limits of taking the last word: it picks up "Road", "Street" and "Kerbside" well, but also fragments such as "Paul's" and "South", so word-based extraction needs checking against the full names.

Q4: String functions: formatting

Question: Which 15 sites had the highest annual mean NO2? Show each as a readable label with its site type in sentence case, its coordinates as "51.523N, 0.155W" and its concentration with units.

Concept: CONCAT, CONCAT_WS, UPPER, LOWER, LEFT, SUBSTRING, FORMAT, ABS

site_label	site_type	location	coordinates	no2_annual_mean	data_capture
GLA4 - Glasgow Kerbside	Urban traffic	Scotland / Central Scotland	55.859N, 4.259W	34.8 ug/m3	97.0%
BEL1 - Belfast Stockman's Lane	Urban traffic	Northern Ireland / Northern Ireland	54.573N, 5.975W	33.5 ug/m3	96.9%
MY1 - London Marylebone Road	Urban traffic	England / Greater London	51.523N, 0.155W	30.6 ug/m3	97.3%
STKR - Stoke-on-Trent A50 Roadside	Urban traffic	England / West Midlands	52.980N, 2.112W	30.6 ug/m3	98.4%
SHBR - Sheffield Barnsley Road	Urban traffic	England / Yorkshire & Humberside	53.405N, 1.456W	29.1 ug/m3	97.3%

site_label	site_type	location	coordinates	no2_annual_mean	data_capture
BDMA - Bradford Mayo Avenue	Urban traffic	England / Yorkshire & Humberside	53.771N, 1.760W	28.7 ug/m3	98.9%
CA1 - Camden Kerbside	Urban traffic	England / Greater London	51.544N, 0.175W	28.6 ug/m3	98.4%
BIRR - Birmingham A4540 Roadside	Urban traffic	England / West Midlands	52.476N, 1.875W	28.2 ug/m3	99.1%
LUTR - Luton A505 Roadside	Urban traffic	England / Eastern	51.892N, 0.462W	27.1 ug/m3	95.3%
LEIR - Leicester A594 Roadside	Urban traffic	England / East Midlands	52.639N, 1.124W	27.0 ug/m3	96.7%
NCA3 - Newcastle Cradlewell Roadside	Urban traffic	England / North East	54.986N, 1.595W	26.9 ug/m3	98.7%
BR11 - Bristol Temple Way	Urban traffic	England / South West	51.458N, 2.584W	25.6 ug/m3	98.6%
DCST - Doncaster A630 Cleveland Street	Urban traffic	England / Yorkshire & Humberside	53.518N, 1.138W	25.5 ug/m3	81.8%
SA33 - Southampton A33	Urban traffic	England / South East	50.920N, 1.463W	25.2 ug/m3	94.3%
DESA - Derby St Alkmund's Way	Urban traffic	England / East Midlands	52.923N, 1.470W	24.9 ug/m3	98.2%

15 rows returned (shown in full).

Interpretation: All 15 sites with the highest annual NO2 are roadside sites, led by Glasgow Kerbside (34.8 µg/m³), Belfast Stockman's Lane (33.5) and London Marylebone Road and Stoke-on-Trent A50 Roadside (both 30.6). Every one is below the UK objective of 40 but roughly 2.5–3.5 times the WHO guideline. Data capture is high (94–99%) except at Doncaster A630 Cleveland Street (81.8%), which still passes the 75% threshold.

Q5: Pattern matching

Question: Do site names that mention a road (Road, Street, Roadside, Kerbside, Lane, Way or an A/M road number) match Defra's official site type, and how does their NO2 compare?

Concept: REGEXP with word boundaries + LIKE

name_pattern	environment_type	sites	sites_with_no2	mean_no2_ugm3
1. Road in name	Urban Traffic	69	53	20.7

name_pattern	environment_type	sites	sites_with_no2	mean_no2_ugm3
2. Centre or Town in name	Urban Background	13	10	15.9
2. Centre or Town in name	Urban Industrial	1	1	11.9
3. Other name	Urban Background	75	49	13.1
3. Other name	Rural Background	24	12	5.0
3. Other name	Urban Traffic	9	7	18.2
3. Other name	Urban Industrial	5	5	13.5
3. Other name	Suburban Background	3	3	9.9
3. Other name	Suburban Industrial	2	2	14.1

9 rows returned (shown in full).

Interpretation: Site names are a reliable sign of a roadside site: all 69 sites whose names mention a road, street, lane, way or A/M road number are officially classed as Urban Traffic, and they average 20.7 µg/m³ of NO2. The reverse is not true, as 9 Urban Traffic sites have other names (average 18.2). Urban background sites average 13.1–15.9, and rural sites 5.0, about a quarter of the roadside level.

Q6: Numeric functions

Question: By region, what is the average annual PM2.5, how many times the WHO guideline (5) is it, and how far is each region from England's 2040 target of 10?

Concept: GREATEST (negatives treated as 0), ROUND, FLOOR, CEIL

region	sites	mean_pm25_ugm3	lowest_site_floor	highest_site_ceil	times_who_guideline	gap_to_2040_target	sites_above_10
Greater London	10	9.16	7	13	1.8	-0.84	2
South East	20	8.56	6	11	1.7	-1.44	2
Yorkshire & Humberside	18	8.52	5	10	1.7	-1.48	0
Eastern	12	8.51	7	12	1.7	-1.49	1
North West & Merseyside	16	8.16	6	10	1.6	-1.84	0
East Midlands	14	8.05	6	10	1.6	-1.95	0
South West	11	7.95	5	10	1.6	-2.05	0
South Wales	5	7.92	6	9	1.6	-2.08	0
West Midlands	14	7.87	6	9	1.6	-2.13	0

region	sites	mean_pm25_ugm3	lowest_site_floor	highest_site_ceil	times_who_guideline	gap_to_2040_target	sites_above_10
North Wales	1	7.38	7	8	1.5	-2.62	0
North East	9	7.20	4	9	1.4	-2.80	0
Northern Ireland	3	6.45	4	8	1.3	-3.55	0
Central Scotland	6	5.73	4	7	1.1	-4.27	0
North East Scotland	1	5.68	5	6	1.1	-4.32	0
Highland	1	4.87	4	5	1.0	-5.13	0

15 rows returned (shown in full).

Interpretation: Greater London has the highest average PM2.5 (9.16 µg/m³, 1.8 times the WHO guideline), followed by the South East, Yorkshire & Humberside and Eastern regions (about 8.5). Every region's average is already below England's 2040 target of 10, but five individual sites are above it (two in London, two in the South East and one in Eastern). Scottish regions are lowest, and Highland (4.87) is the only region at the WHO guideline of 5.

Q7: Conditional functions

Question: Band every site-day of PM2.5 using the UK Daily Air Quality Index (Low 0-35, Moderate 36-53, High 54-70, Very High 71+). How many site-days fall in each band, and how many were at roadside or background sites?

Concept: CASE, IF, NULLIF (safe division)

daqi_band	site_days	pct_of_site_days	roadside_days	background_days	sites_affected	highest_daily_mean
1. Low (DAQI 1-3)	52526	99.43	16935	33205	158	35.5
2. Moderate (DAQI 4-6)	277	0.52	90	176	113	53.2
3. High (DAQI 7-9)	22	0.04	9	11	22	67.7
4. Very High (DAQI 10)	2	0.00	1	1	2	83.5

4 rows returned (shown in full).

Interpretation: 99.43% of the 52,827 PM2.5 site-days were in the Low DAQI band. 277 site-days were Moderate, 22 High and 2 Very High, with a peak daily mean of 83.5 µg/m³. The Moderate days touched 113 different sites, and roadside sites make up about 32% of those days, the same as their share of all days. This suggests PM2.5 episodes are regional events rather than a roadside problem.

Q8: Date functions

Question: At roadside (Urban Traffic) sites, how does NO2 change by day of the week, and how do the morning and evening rush hours compare with the middle of the night?

Concept: DAYOFWEEK, DAYNAME, HOUR

day_no	day_name	readings	all_hours	morning_7_10	evening_16_19	night_1_5	morning_to_night_ratio
1	Sunday	77105	15.4	14.3	19.3	11.3	1.27
2	Monday	76737	19.5	24.7	25.4	10.3	2.40
3	Tuesday	76273	21.0	27.3	26.8	11.2	2.43
4	Wednesday	77673	22.4	28.3	28.5	12.6	2.25
5	Thursday	76447	21.9	28.4	27.5	12.7	2.23
6	Friday	77111	22.1	27.1	27.6	12.7	2.14
7	Saturday	77398	19.6	21.2	24.5	12.9	1.65

7 rows returned (shown in full).

Interpretation: Roadside NO2 follows the working week. Weekday mornings (07:00–10:00) average 24.7–28.4 µg/m³, about 2.1–2.4 times the overnight level, and Wednesday is the highest day overall (22.4). On Sunday the morning peak almost disappears (14.3, only 1.27 times the night level), and the all-day mean is 31% lower than Wednesday's. At weekends the evening is busier than the morning, which reflects leisure rather than commuting traffic.

4. Window Functions (Q9–Q12)

Ranking, distribution, running-window and navigation window functions.

Q9: Ranking window functions

Question: What are the three sites with the highest annual NO2 in each nation, and where does each rank across the UK?

Concept: ROW_NUMBER partitioned by nation + RANK across the UK

nation	nation_rank	site_name	environment_type	no2_ugm3	uk_rank
England	1	London Marylebone Road	Urban Traffic	30.6	3
England	2	Stoke-on-Trent A50 Roadside	Urban Traffic	30.6	4
England	3	Sheffield Barnsley Road	Urban Traffic	29.1	5
Northern Ireland	1	Belfast Stockman's Lane	Urban Traffic	33.5	2
Northern Ireland	2	Armagh Roadside	Urban Traffic	21.2	29

nation	nation_rank	site_name	environment_type	no2_ugm3	uk_rank
Northern Ireland	3	Derry Rosemount	Urban Background	8.1	126
Scotland	1	Glasgow Kerbside	Urban Traffic	34.8	1
Scotland	2	Aberdeen Wellington Road	Urban Traffic	22.1	24
Scotland	3	Edinburgh Nicolson Street	Urban Traffic	20.7	32
Wales	1	Chepstow A48	Urban Traffic	22.3	23
Wales	2	Hafod-yr-ynys Hill Roadside	Urban Traffic	20.2	34
Wales	3	Cardiff Newport Road	Urban Traffic	18.3	45

12 rows returned (shown in full).

Interpretation: Glasgow Kerbside (34.8 µg/m³) and Belfast Stockman's Lane (33.5) are the two highest NO2 sites in the UK, ahead of England's leaders, London Marylebone Road and Stoke-on-Trent A50 Roadside (30.6 each). Wales's highest site, Chepstow A48 (22.3), ranks only 23rd in the UK. Northern Ireland has just three NO2 sites with a full year, so its third place goes to an urban background site at 8.1.

Q10: Distribution window functions

Question: Split PM2.5 sites into four equal groups by annual mean. What range does each quartile cover, what percentile does it start at, and what mix of site types is in each?

Concept: NTILE, PERCENT_RANK, CUME_DIST

quartile	sites	min_pm25	max_pm25	avg_pm25	starts_at_percentile	cumulative_pct	roadside	urban_background	rural_background	other
1	36	4.38	7.47	6.46	0	25.4	9	17	9	1
2	36	7.51	8.10	7.81	25.5	50.7	4	24	6	2
3	35	8.10	8.77	8.45	51.1	75.4	12	19	2	2
4	35	8.79	12.27	9.48	75.9	100	20	11	1	3

4 rows returned (shown in full).

Interpretation: Annual PM2.5 falls in a narrow range: the middle half of the 142 sites lies between 7.51 and 8.77 µg/m³, and the highest site is 12.27. Roadside sites cluster at the top, with 20 of the 35 sites in the top quartile classed as Urban Traffic, while the bottom quartile (4.38–7.47) holds 9 of the 18 rural sites. Even the bottom quartile averages 6.46, so almost every site is above the WHO guideline of 5.

Q11: Window aggregate functions

Question: Using the 8-hour running mean of ozone, which 15 sites had the most days with a daily maximum above 100, and which met the UK objective (no more than 10 such days)?

Concept: AVG() and COUNT() OVER a time-based RANGE window (the current hour and the 7 before it)

site_name	environment_type	nation	days_with_data	days_above_100	highest_8h_mean	uk_objective
Milton Keynes Civic Centre	Urban Background	England	365	36	158.5	Not met
London Westminster	Urban Background	England	361	31	170.0	Not met
London Bloomsbury	Urban Background	England	365	31	163.7	Not met
Derby Stockbrook Park	Urban Background	England	365	31	154.0	Not met
Wicken Fen	Rural Background	England	316	31	148.6	Not met
London Honor Oak Park	Urban Background	England	360	30	161.1	Not met
Yarner Wood	Rural Background	England	360	30	159.9	Not met
Leicester University	Urban Background	England	365	29	156.7	Not met
Cwmbran Crownbridge	Urban Background	Wales	365	29	152.1	Not met
Bournemouth	Urban Background	England	323	29	152.0	Not met
Tallington	Rural Background	England	359	29	149.6	Not met
London N. Kensington	Urban Background	England	361	28	164.5	Not met
Nottingham Kenmore Gardens	Urban Background	England	365	28	156.4	Not met
Sibton	Rural Background	England	365	27	164.6	Not met
Leamington Spa	Urban Background	England	356	27	151.0	Not met

15 rows returned (shown in full).

Interpretation: All 15 sites with the most high-ozone days failed the UK objective, with 27–36 days above 100 µg/m³ against an allowance of 10. Milton Keynes Civic Centre had the most (36), and London Westminster recorded the highest 8-hour mean (170.0). Every one is an urban or rural background site, none roadside, because traffic emissions of nitric oxide remove ozone close to roads. High ozone is therefore a background issue, different in kind from NO2.

Q12: Navigation window functions

Question: Which 15 days saw the biggest day-on-day rise in the network-wide average PM2.5, and what happened the day before and the day after?

Concept: LAG and LEAD over a daily series

obs_date	day_name	sites	prev_day_pm25	pm25	next_day_pm25	rise_from_prev_day
2025-06-09	Monday	149	5.6	19.9	5.8	14.3
2025-03-09	Sunday	135	18.8	32.7	40.4	13.9
2025-03-19	Wednesday	139	11.4	22.6	23.1	11.1
2025-02-08	Saturday	135	8.8	18.5	22.4	9.7
2025-07-23	Wednesday	146	3.6	12.6	13.1	9.0
2025-03-23	Sunday	140	15.2	23.0	25.2	7.8
2025-03-10	Monday	136	32.7	40.4	7.7	7.7
2025-02-15	Saturday	133	17.3	24.9	18.6	7.7
2025-04-03	Thursday	144	9.6	17.1	16.4	7.6
2025-01-11	Saturday	130	16.1	22.7	19.7	6.6
2025-02-19	Wednesday	134	22.7	29.0	6.8	6.2
2025-04-26	Saturday	142	9.9	16.1	10.3	6.2
2025-02-18	Tuesday	134	16.5	22.7	29.0	6.2
2025-01-08	Wednesday	132	4.2	9.7	11.2	5.5
2025-11-05	Wednesday	153	4.7	10.2	13.2	5.5

15 rows returned (shown in full).

Interpretation: The largest one-day rise in network-average PM2.5 was on 9 June 2025 (5.6 to 19.9 µg/m³), which fell back to 5.8 the next day: a short, sharp event. The most sustained episode was 8–10 March, when the average climbed from 18.8 to 32.7 and then 40.4 before dropping to 7.7. Eight of the 15 days fall in February and March, consistent with the late-winter and spring particulate episodes typical of the UK. Bonfire Night shows up too, with 5 November rising from 4.7 to 10.2 and 13.2 the following day.

5. Subqueries, Set Operations and Statistics (Q13–Q15)

Correlated subqueries with EXISTS, set operations and statistical aggregates.

Q13: Subqueries

Question: Which roadside sites have an annual mean NO2 above the roadside average for their own nation, and does the same site also record PM2.5 above England's 2040 target of 10?

Concept: Correlated subquery (nation average per row) + EXISTS against the PM2.5 results for the same site

site_name	nation	no2_ugm3	nation_roadside_avg	pm25_above_10
Glasgow Kerbside	Scotland	34.8	19.0	Not measured
Belfast Stockman's Lane	Northern Ireland	33.5	27.4	Not measured
London Marylebone Road	England	30.6	20.5	Yes
Stoke-on-Trent A50 Roadside	England	30.6	20.5	No
Sheffield Barnsley Road	England	29.1	20.5	No
Bradford Mayo Avenue	England	28.7	20.5	Not measured
Camden Kerbside	England	28.6	20.5	Yes
Birmingham A4540 Roadside	England	28.2	20.5	No
Luton A505 Roadside	England	27.1	20.5	No
Leicester A594 Roadside	England	27.0	20.5	No
Newcastle Cradlewell Roadside	England	26.9	20.5	No
Bristol Temple Way	England	25.6	20.5	No
Doncaster A630 Cleveland Street	England	25.5	20.5	No
Southampton A33	England	25.2	20.5	No
Derby St Alkmund's Way	England	24.9	20.5	No
Haringey Roadside	England	24.5	20.5	Not measured
Widnes Milton Road	England	24.2	20.5	Not measured

site_name	nation	no2_ugm3	nation_roadside_avg	pm25_above_10
Oldbury Birmingham Road	England	24.0	20.5	Not measured
St Helens Linkway	England	23.6	20.5	Not measured
Chepstow A48	Wales	22.3	18.8	No

20 rows returned (shown in full).

Interpretation: Glasgow Kerbside's NO2 (34.8 µg/m³) is 83% above the Scottish roadside average (19.0). England's highest roadside sites are up to 49% above the English roadside average (20.5). Of these 20 high-NO2 sites, only London Marylebone Road and Camden Kerbside also have PM2.5 above 10, and 11 are below it. Seven, including Glasgow Kerbside and Belfast Stockman's Lane, have no full year of PM2.5 data, a monitoring gap at some of the most polluted roadsides.

Q14: Set operations

Question: Compare the 20 sites with the highest annual NO2 with the 20 with the highest annual PM2.5. Which sites are in both lists, and which in only one?

Concept: UNION ALL of two ranked lists + GROUP BY to classify

appears_in	site_name	environment_type	no2_ugm3	pm25_ugm3
1. Both lists	London Marylebone Road	Urban Traffic	30.6	11.0
1. Both lists	Sheffield Barnsley Road	Urban Traffic	29.1	9.1
1. Both lists	Camden Kerbside	Urban Traffic	28.6	12.3
1. Both lists	London Hillingdon	Urban Background	24.8	9.4
2. High NO2 only	Glasgow Kerbside	Urban Traffic	34.8	NULL
2. High NO2 only	Belfast Stockman's Lane	Urban Traffic	33.5	NULL
2. High NO2 only	Stoke-on-Trent A50 Roadside	Urban Traffic	30.6	8.9
2. High NO2 only	Bradford Mayo Avenue	Urban Traffic	28.7	NULL

appears_in	site_name	environment_type	no2_ugm3	pm25_ugm3
2. High NO2 only	Birmingham A4540 Roadside	Urban Traffic	28.2	8.7
2. High NO2 only	Luton A505 Roadside	Urban Traffic	27.1	9.0
2. High NO2 only	Leicester A594 Roadside	Urban Traffic	27.0	9.0
2. High NO2 only	Newcastle Cradlewell Roadside	Urban Traffic	26.9	7.3
2. High NO2 only	Bristol Temple Way	Urban Traffic	25.6	9.0
2. High NO2 only	Doncaster A630 Cleveland Street	Urban Traffic	25.5	8.9
2. High NO2 only	Southampton A33	Urban Traffic	25.2	8.9
2. High NO2 only	Derby St Alkmund's Way	Urban Traffic	24.9	8.8
2. High NO2 only	Haringey Roadside	Urban Traffic	24.5	NULL
2. High NO2 only	Widnes Milton Road	Urban Traffic	24.2	NULL
2. High NO2 only	Oldbury Birmingham Road	Urban Traffic	24.0	6.5
2. High NO2 only	St Helens Linkway	Urban Traffic	23.6	NULL
3. High PM2.5 only	Manchester Piccadilly	Urban Background	23.6	9.6
3. High PM2.5 only	Hull Holderness Road	Urban Traffic	20.7	9.5
3. High PM2.5 only	Leeds Headingley Kerbside	Urban Traffic	20.2	9.8
3. High PM2.5 only	Salford Eccles	Urban Background	18.0	9.8

appears_in	site_name	environment_type	no2_ugm3	pm25_ugm3
3. High PM2.5 only	Hull Freetown	Urban Background	17.4	9.1
3. High PM2.5 only	Stanford-le-Hope Roadside	Urban Traffic	16.9	11.3
3. High PM2.5 only	London Bexley	Suburban Background	16.4	9.7
3. High PM2.5 only	Chatham Roadside	Urban Traffic	15.2	10.0
3. High PM2.5 only	Nottingham Kenmore Gardens	Urban Background	13.9	9.4
3. High PM2.5 only	Christchurch Barrack Road	Urban Traffic	13.8	9.7
3. High PM2.5 only	Rochester Stoke	Rural Background	10.7	9.6
3. High PM2.5 only	Maidstone Upper Stone Street	Urban Traffic	NULL	10.2
3. High PM2.5 only	Rotherham East Dene	Urban Background	NULL	9.9
3. High PM2.5 only	Worthing East Ten Acres	Urban Background	NULL	9.8
3. High PM2.5 only	Slough Windmill Bath Road	Urban Traffic	NULL	9.6
3. High PM2.5 only	Bradford Shipley Airedale Road	Urban Traffic	NULL	9.1

36 rows returned (shown in full).

Interpretation: Only four sites are in the top 20 for both pollutants: London Marylebone Road, Sheffield Barnsley Road, Camden Kerbside and London Hillingdon. The 16 high-NO2-only sites are all roadside, and six of them have no PM2.5 annual value. The 16 high-PM2.5-only sites are a mix of background and roadside sites, mostly in the north of England and the South East, and five have no NO2 monitor. NO2 is driven by local traffic, while PM2.5 is more regional.

Q15: Statistical aggregate functions

Question: At sites measuring both NO2 and ozone, how variable is each pollutant hour to hour, and how strongly are the two correlated, by site type?

Concept: AVG, STDDEV_POP, VAR_POP + Pearson correlation (covariance / product of standard deviations)

environment_type	sites	paired_hours	mean_no2	sd_no2	var_no2	mean_o3	sd_o3	var_o3	correlation_r
Urban Background	59	464054	13.7	12.5	155.1	52.7	23.2	540.4	-0.593
Rural Background	15	111307	5.3	6	36.1	59.5	20.5	420.2	-0.402
Urban Industrial	3	25712	14.8	13.5	183.1	53.9	24	578.1	-0.601
Urban Traffic	3	22398	26.9	17.1	292.4	40.9	21.9	478.6	-0.58
Suburban Background	2	16624	6.8	7	49	53.1	23.7	560.3	-0.505
Suburban Industrial	1	8047	13.8	12.1	147.2	51.8	24.5	599.6	-0.597

6 rows returned (shown in full).

Interpretation: NO2 and ozone are negatively correlated at every site type (r from -0.40 to -0.60): when NO2 is high, ozone is low. This is the chemistry seen in Q11, where nitric oxide from traffic converts ozone into NO2. The link is strongest at urban background and industrial sites (about -0.59 to -0.60) and weakest at rural sites (-0.40). Ozone is more variable than NO2 (variance 420–600 against 36–292). Only 3 roadside sites measure ozone at all, which shows that ozone is monitored mainly away from roads.

6. Summary of Findings

For a local authority air quality or public health officer, 2025 shows the UK meeting its legal annual objectives everywhere but still well short of health-based guidelines, with five priorities:

1. **NO2 is a roadside problem.** No site exceeded the UK objective of 40 µg/m³, but 115 of 142 sites (81%), including 58 of 60 roadside sites, were above the WHO guideline of 10 (Q2). The 15 highest sites are all roadside, led by Glasgow Kerbside (34.8) and Belfast Stockman's Lane (33.5) (Q4, Q9). Roadside NO2 on weekday mornings is more than twice the overnight level, and Sunday is 31% lower than Wednesday (Q8), so traffic measures at peak times target the main source.
2. **PM2.5 is regional and close to the 2040 target.** Every region averages below England's 2040 target of 10, with Greater London highest at 9.16, but only Highland reaches the WHO guideline of 5 (Q6). The middle half of sites lies within 7.5–8.8 µg/m³ (Q10), and only four sites are in the top 20 for both NO2 and PM2.5 (Q14). PM2.5 needs regional action, not just local traffic schemes.

3. **Episodes are rare but widespread.** 99.4% of PM2.5 site-days were Low on the Daily Air Quality Index (Q7). The worst episode ran from 8 to 10 March, when the network average reached 40.4, and Bonfire Night stands out as well (Q12). Moderate days hit 113 sites at once, so public alerts should be regional.
4. **Ozone fails the UK objective at background sites.** The 15 worst sites each had 27–36 days above the 8-hour threshold against an allowance of 10, and all were urban or rural background sites (Q11). NO2 and ozone are negatively correlated at every site type (r = -0.40 to -0.60) (Q15), so cutting NO2 may raise ozone locally.
5. **There are monitoring gaps at the most polluted roadsides.** Seven of the 20 roadside sites furthest above their nation's average, including the two highest in the UK, have no full year of PM2.5 data (Q13). Only 3 roadside sites measure ozone (Q15).

Limitations:

- The data covers one year, so trends cannot be measured.
- Annual means use sites with at least 75% data capture, and the network covers a few hundred locations, not every street.
- Site names were parsed by word position, which is imperfect (Q3).
- Extreme spikes such as the PM10 maximum (Q1) were not removed.
- The benchmarks are applied to individual monitoring sites, which is not the same as a formal compliance assessment.